KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

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UNIT ONE

TOPIC: INTRODUCING FOREST ENTOMOLOGY

Objectives: by the end of this unit, SSBAT;

- Explain the scope of entomology
- Explain the Principles of entomology
- Explain the relationship between entomology and pathology
- Explain Significance of entomology to agriculture

Introduction to entomology

Entomology is the study of insects, but often includes closely related arthropodsas well.Entomology is divided into basic and applied entomology. The former deals with the basic aspects of insects, whereas the latter deals with the economic aspects of insects in human society.

Other arthropods have a more reduced type of body segmentationsuch as arachnids (spiders, scorpions and mites), myriapods (millipedes and centipedes) and even crustaceans (crabs and

isopods). Nevertheless, insects (organisms belonging to the class Insecta) are studied by entomologists, arachnids are studied by arachnologists and acarologists (those who specialize on mites and ticks), myriapods are studied by myriapodologists, and crustaceans are studied by carcinologists (also known as malacostracologists or crustaceologists). The work of entomologists (people who study insects) is extremely diverse due to the high abundance, species richness, and ecological and behavioral characteristics of insects.

Agricultural Entomology

Forest entomology is the study of insects and other arthropods associated with ecosystems.These insects or arthropods could be of economic importance to forests by their feeding habits.Hence any insects that feeds on forests could be a potential pest.According to forest entomology, an insect is considered a pest when it is harmful to plant or human life, be it directly or indirectly.

Various insects play different roles in the environment or forest ecosystems. An insect may be beneficial or detrimental depending on the circumstances. Some beneficial insects include pollinators and natural enemies while destructive types include defoliators and all types of borers. Damage caused by borers may include perforation of tree bark, leaves and roots, which often become an entry route for pathogens (fungi, virus, others); crown dieback which consists of a substantial progressive decline in crown health, often resulting in tree death; degradation of wood quality through tunneling caused by insect feeding; staining of the wood either by feeding or by fungi carried by the tunneling insect; destruction of flowers and seeds; and decrease in photosynthetic activity and other physiological processes due to defoliation or injection of toxins.

On the other hand, in general, insect populations in forest ecosystems are in balance. But insects are also essential in maintaining healthy forests. Some services provided by insects in a forest include the aeration of soil through tunneling activity which improves gas exchange in the root system; decomposition of organic matter of the forest floor; pollination; biological control; food source for other invertebrates and vertebrates that live in the forest; and production of food products such as honey.

Scope of entomology

Entomology may include studies on biodiversity, biology and ecology of insects in natural or cultivated forest ecosystems, and damage assessment to tree structures, forest stands and wood products. It deals mostly with insect pest management, seeks to control insects that cause crown dieback or death of trees, degradation and destruction of wood, defoliation and other problems related to the health of a forest and wood products.
Agricultural entomological problems can no longer be disregarded. They must be faced squarely and solved together with problems in agriculture , management and utilization.

It is important, therefore, that every forester should be able to recognize evidences of possibly dangerous insect activities and should know enough about insect pests and their control to approach problems intelligently and determine which insects are beneficial, and which are neutral in their relation to the life history of the plant and the future usefulness of its products.

You should know how and where to obtain information about insects and should be able to apply the necessary remedies intelligently.

Principles of Entomology

Agricultural entomology embraces only such insects that are in some way related to human interest in forests and forest products. Therefore, forest entomology will be approached through consideration of general underlying principlesfrom the forestry viewpoint. The emphasis is placed on the influence of insects on the forest rather than on the insects themselves.

The objective of forest entomology is to regulate insect numbers in such a manner that their activities do not cause economically important losses and that control measures are tailored to current and future forest practices as closely as possible. This ecological and practical approach seems essential if the principles of forest entomology are to be understood and applied by foresters.

Relationship between Entomology and Pathology

Entomology and pathology both involves biotic organisms and their direct and indirect influence on forests, and their consequential interactions in the forest ecosystem is quite enormous. Certain insect species for example, may cause indirect damage to forests by spreading diseases and these are termed as vectors. In other instances by direct damage through perforations of tree bark, leaves and roots, which often become an entry route for pathogens such as fungi, virus and others.

Significance of entomology to natural resource management

Insects often affect management of natural resources such as trees in the following ways:

- Some insect groups attack flowers and seeds of crops, even before the seed can be gathered.
- Insects invade seeds where they are stored and after they are planted in nurseries
- Some insects infest roots, stems and foliage of seedlings.
- In natural forests or plantations, insects defoliate, bore and live in the bark or wood.
- Others live in roots and some transmit pathogens.

To fully appreciate insect diversity and to work in applied area of protecting forest investments and ecosystems, it is important to understand basic entomology so as to mitigate the economic losses that insects may pose to forests.

The effects of insects on the ecosystem

The economic impact of insects is measured not only by the market value of products they destroy and the cost of damage they inflict but also by the money and resources to be expended

on prevention and control of the pest outbreak. Other effects of insects can be ecological, social and cultural. The following are some which can fall in any one of the aforesaid:

1.2.1. Harmful insects

Some insects are harmful to plants and animals; these carry plant and animal diseases, destroy crops and forests and are considered insect pests, including wood boring insects, defoliators, seed insects, tsetse flies etc.

1.2.2. Scavenging insects

Scavengers are insects or organisms that feed primarily on dead plants or animals, decaying vegetation or animal faeces. The food for scavengers is not limited to dead material because microbes such as bacteria and fungi flourish on food consumed by scavengers. Scavengers are specific to either plant or animal materials. Many species of insects feed on various kinds of dead plants and animal material including playing a role in the recycling of nutrients. Many soil animals (insects and other arthropods, such as earthworms) facilitate the oxygenation of the soil through burrowing into the soil and also enables the movement of capillary water in the soil.

1.2.3. Insects that destroy plants and products

There are many insects that destroy or attack economically important plants. Some insects attack various products to include wood and stored seeds. The storage losses caused by insects can be greater than losses sustained while growing the crops.

1.2.4. Insects important for food and products

Insects are a source of food for many animals to include birds, fish, mammals and reptiles. Insects are rich in proteins and lipids necessary for the rapid growth of young birds. Immature and adult insects are important food source for fresh water fish. Insects have been eaten by human beings for thousands of years as important sources of protein and fat. Many insects such as termites and grasshoppers contain a rich source of protein, fat and vitamins.

1.2.5. Insects as pollinators

Sexual reproduction in flowering plants occurs when wind or insects transfer pollen from the anther of a male flower to a stigma of a female flower. Many species of trees are insect pollinated and include species of cereals such as corn and wheat. Most fruit trees and many

ornamental flowers and field crops depend on insects for pollination. Most insect pollinators are found in the orders **Lepidoptera (butterflies), Hymenoptera (bees) and Coleoptera (beetles).** Above all, insect pollination is an important and complex subject that covers many ecological and evolutionary principles.

1.2.6. Insects as natural Enemies

Most insect species rarely reach epidemic population levels because they are kept at endemic levels by natural enemies. Biological control refers to the use of natural enemies either native or exotic to regulate insect pest populations. Natural enemies include all predators, parasites and pathogens of insects. Biological control often covers two major areas:

1. Natural biological control: work of natural enemies without human involvement
2. Applied biological control is the manipulation of natural enemies for pest suppression

Entomology enable agro foresters to appreciate the roles insects play in ecosystem and hence to properly mitigate the bad effects of insects so as to save the ecosystem through enhancement of the positive aspects of the insects and by use of environmentally sound control and preventive measures of insects in order to save forest resources under insect destruction and the ecosystem.